POWER PULSE

Economic Load Dispatch Important Formulae

Notes by Balaji Sir

1 Without losses

1.

$$F_i = a_i P_i^2 + b_i P_i + c_i \qquad IC_i = \frac{dF_i}{dP_i} = 2a_i P_i + b_i$$

₹/h and ₹/MWh

2.

$$\frac{dF_1}{dP_1} = \frac{dF_2}{dP_2} = \cdots = \lambda, \qquad \sum P_i = P_D$$

equal incremental cost

3.

$$P_i = \frac{\lambda - b_i}{2a_i}, \qquad \lambda = \frac{P_D + \sum \frac{b_i}{2a_i}}{\sum \frac{1}{2a_i}}$$

4.

$$\frac{dF}{dP} = \lambda \ (P_{min} < P < P_{max}); \ \ \leq \lambda \ (P = P_{max}); \ \ \geq \lambda \ (P = P_{min})$$

with generation limits

5.

$$\Delta C \approx \lambda \, \Delta P_D$$

cost of an extra MW at the load

2 With transmission losses

6.

$$P_L = \sum_i \sum_j P_i B_{ij} P_j, \qquad \frac{\partial P_L}{\partial P_i} = 2 \sum_j B_{ij} P_j$$

B-coefficients (MW^{-1})

7.

$$\sum P_i = P_D + P_L$$

8.

$$\frac{dF_i}{dP_i} + \lambda \frac{\partial P_L}{\partial P_i} = \lambda \ \Rightarrow \ \frac{dF_i}{dP_i} = \lambda \left(1 - \frac{\partial P_L}{\partial P_i}\right)$$

coordination equation

9.

$$PF_i = \frac{1}{1 - \partial P_L / \partial P_i}, \qquad PF_i \, \frac{dF_i}{dP_i} = \lambda$$

penalty factor

3 Remember

10.

$$\text{Unit commitment} \neq \text{economic dispatch}$$

UC decides which units are on (hours/days); ED shares the load among units already on

11.

$$\text{Saving} = C(\text{equal sharing}) - C(\text{optimum})$$